CS6735 Machine Learning & Data Mining
Primary Report

FS6: Relief / ReliefF

Group Members:
Aadesh Kumar (3809131)
Sudhish Poojary (3711002)

Table of Contents

1. Introduction and Objectives **2**

2. Importance of Feature Selection **2**

3. Relief / ReliefF Algorithm **3**

3.1 Algorithm 3

4. Implementation **4**

4.1 Datasets 5

5. References **6**

1. Introduction and Objectives

Feature selection methods are generally categorized into different methods that include filter, wrapper, and embedded approaches. Many literature papers support the claim that there is no universal "best method" for all tasks [1]. However, filter methods are considered particularly attractive because they are computationally efficient and independent of any specific learning algorithm. One widely studied feature-based method is the ReliefF algorithm, which evaluates feature importance based on how well feature values differentiate between different instances that are neighbouring each other.

The objective of this project is to evaluate how feature selection affects the performance of multiple classification models across multiple datasets. Specifically, we investigate the impact of the **Relief / ReliefF** feature selection algorithm on classification accuracy and F1-score.

For each dataset, the proposed experiment will be conducted in two phases:

1. **Baseline Phase** – Training classifiers using the full feature set.
2. **Feature Selection Phase** – Applying ReliefF to select relevant features, then retraining the classifiers.

A total of five classifiers will be evaluated under a 10-fold cross-validation framework with hyperparameter tuning to ensure fair comparison.

2. Importance of Feature Selection

Increasingly large datasets lead to data with very high dimensionality, where the number of features can be large relative to the number of instances. In many such cases, only a subset of the available features is relevant to predictions. Irrelevant features tend to add noise, increase computational burden, and make it more difficult for learning algorithms to identify meaningful patterns. Increase in dimensionality leads to modelling methods suffering from problems such as overfitting, and increased training time. Feature selection helps in identifying the relevant features and discarding irrelevant ones [2]. Therefore, selecting an appropriate feature selection method is a critical step in a successful data mining pipeline.

3. Relief / ReliefF Algorithm

Relief-Based Algorithms (RBAs) are the only individual evaluation filter algorithms capable of detecting feature dependencies. Other filter-based algorithms do not inherently search through feature combinations, but rather use the concept of nearest neighbors to derive feature statistics that indirectly account for interactions [2].

RBAs also tend to retain the generalized filter algorithms as well, i.e. they are relatively fast (with an asymptotic time complexity of $O(instances^2 \cdot features)$) and selected features with different induction algorithm dependent [2]. Therefore, in summary unlike wrapper and embedded approaches, RBAs:

- Do not rely on a specific classifier
- Are computationally efficient
- Are model-agnostic

3.1 Algorithm

Since Relief is an individual evaluation filtering feature selection method, it calculates a proxy statistic for each feature that can be used to estimate feature 'quality' or 'relevance' to the target concept. These statistics are referred as feature weights ($W[A]$ = weight of feature 'A'), or feature 'scores' that can range from -1 (worst) to +1 (best). The original Relief algorithm is shown in figure 1 below [2]:

Algorithm 1 Pseudo-code for the original Relief algorithm

```
Require: for each training instance a vector of feature values and the class value
  n ← number of training instances
  a ← number of features (i.e. attributes)
  Parameter: m ← number of random training instances out of n used to update W

  initialize all feature weights W[A] := 0.0
  for i:=1 to m do
    randomly select a 'target' instance R_i
    find a nearest hit 'H' and nearest miss 'M' (instances)
    for A:= 1 to a do
      W[A] := W[A]−diff(A, R_i, H)/m+diff(A, R_i, M)/m
    end for
  end for
  return the vector W of feature scores that estimate the quality of features
```

Figure 1: Pseudo-code for the original Relief Algorithm [2]

For each of m randomly selected training instances R_i (chosen without replacement), the Relief algorithm R_i as the target instance and updates the feature weight vector W based on local differences.

In each iteration:

- The distance between the target instances R_i and all other training instances is computed
- The algorithm identifies:
  - **Nearest Hit (H)**: closest instance from the same class
  - **Nearest Miss (M):** closest instance from a different class
- For each feature A, the weight $W[A]$ is updated according to the difference in feature values between the target instances and the nearest hit/miss.

Features that differ between R_i and the nearest miss M increase their weight, whereas the features that differ between R_i and the nearest miss H decrease their weight.

ReliefF extends the original Relief algorithm by:

- Handling multi-class problems
- Using multiple nearest neighbors
- Providing more robust feature weight estimation

In this project, we will be using the **`skrebate.ReliefF`** implementation for feature selection.

4. Implementation

The project consists of two experimental phases:

Phase 1 (Baseline Classification): All five classifiers (listed in the classifiers section in the Projects List document) will be trained using the full feature dataset. Their performance will be recorded to establish baseline results.

Phase 2 (ReliefF Feature Selection): Within each fold of 10-fold-cross-validation, ReliefF will be applied to the training data to compute feature relevance scores. A subset of top-ranked features will be selected, and the classifiers will be retrained using only these features. Feature selection will be performed exclusively on the training data to prevent data leakage.

The performance of ReliefF will be evaluated using Accuracy and F1-score. Results will be presented as mean ∓ standard deviation across the 10 folds.

4.1 Datasets

There are two main datasets that we can validate the ReliefF algorithm with along with the dataset that will be provided:

1. **UCI Breast Cancer Wisconsin** – Contains 30 numerical features, providing sufficient dimensionality for meaningful feature ranking.
2. **UCI Wine Dataset** – Contains 13 numerical features derived from chemical analysis and is a multi-class classification making it appropriate to test the robustness of the ReliefF algorithm since it is known to be better than the original for multi-class problems.

5. References

[1] V. Bolón-Canedo, N. Sánchez-Maroño, and A. Alonso-Betanzos, "A review of feature selection methods on synthetic data," *Knowl. Inf. Syst.*, vol. 34, Mar. 2012, doi: 10.1007/s10115-012-0487-8.

[2] R. J. Urbanowicz, M. Meeker, W. LaCava, R. S. Olson, and J. H. Moore, "Relief-Based Feature Selection: Introduction and Review," Apr. 02, 2018, *arXiv*: arXiv:1711.08421. doi: 10.48550/arXiv.1711.08421.